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Sex-differences in the Agonistic Behavior of Spiny-mice (*Acomys cahirinus*)

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Abstract

Observations of 8 groups each containing three adult spiny mice (*Acomys cahirinus*) revealed that instances of chasing and physical displacement were quite common, while fighting and biting were rarely observed. The relationships between the most frequent behavioral categories were presented in a correlation matrix.

In a second study, ♀♀ tested in their home cages were dominant over ♂♂. In the ♂♂' home cages, however, no differences in the frequency of ♀ vs. ♂ aggression were observed. More instances of aggression were observed in the cages of the ♀♀ than in the cages of the ♂♂.

Introduction

Extensive studies of rodent agonistic behavior have focused primarily upon the common laboratory rat (*Rattus norvegicus*) and mouse (*Mus musculus*), with a considerably lesser number of research efforts being devoted to such other common species as the golden hamster (*Mesocricetus auratus*) and Mongolian gerbil (*Meriones unguiculatus*). The social interactions and general behavior of the remaining species of the order Rodentia (which includes the largest number of species and individuals among the class Mammalia) (DAVIS and GOLLEY 1963) have been either little studied or entirely neglected. The differences in agonistic behavior patterns among the more commonly investigated species certainly suggest that research into the behavior of other species of rodents (especially the less domesticated species) would be worthwhile.

The present report concerns social interactions and agonistic behavior in *Acomys cahirinus* (spiny mouse), a murid rodent which is indigenous to the Near East, and breeds readily in captivity. To date, behavioral studies of *A. cahirinus* have been primarily concerned with the precocial offspring (e. g. PORTER and ETSCORN 1974; PORTER and RUTTLE 1975; RUCH 1967; DIETERLEN 1962) — with little research being reported relating to adult social behavior.

I. A Descriptive Study of Social Interactions in Triads of *Acomys cabirinus*

Methods

Adult *Acomys cabirinus* which had been born and raised in the laboratory colony were used in this study. On the day preceding the beginning of the observations, the animals who were to make up a given triad were placed into the observation room in individual cages. The lighting schedule in this room was opposite to that of the colony room; with the overhead lights going on at 18:00 and off at 07:00 each day. A 40-watt fluorescent lamp oriented away from the observation cage was on constantly to provide the minimal illumination necessary to observe the animals during the "dark" phase of the cycle. It must be pointed out that the abrupt change in the illumination cycle may be expected to influence the absolute rate of behavior of the subjects for several days afterwards. However, since the current study is primarily concerned with describing agonistic interactions, as well as comparing the relative rates of such behavior patterns between the sexes, the changed illumination cycle should not have been a critical factor.

Observations began on the day following the isolation of the subjects in the observation room. At that time, the three mice which would comprise a given triad were introduced together into a large cage, where they remained throughout the study. By observing groups consisting of three animals, the likelihood of noting instances of overt social interactions should be fairly high since the possibility of simultaneous interactions among more than one pair (or triad) of animals is precluded. Whenever possible, naturally occurring anatomical features (e.g., the tail being short or missing) were used for individual identification of subjects. When identification in this manner was not possible, small areas on the backs or tails of the mice were marked with spray paint.

The observation cage consisted of two $34 \times 34 \times 24$ inch high sections which were bolted together to form one $68 \times 34 \times 24$ inch high cage with a wire-mesh barrier between the two halves. A 4-inch diameter tunnel which allowed access into both compartments was situated in the center of this partition one inch above floor level. One side of the cage contained a coffee cup and food tray, and the other side of the cage contained a large wooden block. The cage was otherwise empty except for a thick layer of bedding material (sugar-cane waste).

Observation periods were 30 min long with the first commencing with the introduction of the animals into the observation cage. A given group of animals was observed for two 30-min periods each day Mon.—Fri., with single 30-min sessions on Saturday and Sunday. On Mon.—Fri., one observation session was held in the morning (i.e., beginning at 09:00—10:00), and the second in the early afternoon. All observations were made during the dark phase of the illumination cycle, with the observer sitting quietly next to the cage.

Each triad was observed for 20 consecutive days, or until one of the animals died or was killed. At the conclusion of the observations for one triad (i.e., after 20 days maximum), the observation-cage was cleaned in preparation for introduction of the next triad. A total of eight triads comprising two replications of each of the four possible sex combinations (e.g., all ♂♂; 2 ♂♂ and 1 ♀) were observed during the course of this study. The initial triads of all four possible sex combinations were comprised of animals that had been housed in heterosexual pairs after weaning and were isolated several days prior to the beginning of the observation period. The animals in the second triads (replications) of each of the four sex-combinations had been reared in isolation following weaning at 30 days of age. In no instances had any of the animals in a triad been exposed to one another prior to being placed into the observation cage.

Although food was available ad lib. during the time that the animals spent in the observation-cage, water was never present. Moisture was obtained from apple chunks which were dropped into the cage at the end of the first 30-min session each day. At that time, a single piece of apple was dropped into the cage (thereby establishing a food competition situation) and the animals observed for 5-min to determine which one(s) successfully obtained and ate the apple. Further apple pieces were dropped into the cage following this 5-min period.

Behavioral categories

For the first triad, all readily observable social interactions (i.e., gross motor patterns occurring when at least two of the three animals were in physical proximity to one another, or when one animal moved towards another) that occurred during the 30-min session were recorded. Following several hours of such observations and note taking, it became apparent that a small number of broadly defined behavioral categories tended to recur consistently and account for a large proportion of the descriptive data. Therefore, a behavioral checklist was designed to facilitate subsequent data collection. For each of the major behavioral categories (identified from the initial observations), a matrix was drawn enabling the observer

to score frequencies of occurrence of that behavior according to which animal performed that pattern as well as the animal towards which (or in whose presence) it was directed. Total occurrences of the following behavior patterns were scored in this manner:

Approaches — One animal was scored as approaching a second when it moved towards and came within approximately 1 foot of the second while the second animal was either stationary or moving slowly about the cage.

Chases — During chase, one animal closely follows behind a second animal who is rapidly running. It appears as if the animal being "chased" is attempting to lose the pursuing animal.

Displaces — Animal A is said to "displace" B when B moves away from an Approach by A, and A moves into the area previously occupied by B. Unlike "chase", A does not then continue to follow the "displaced" animal. Thus, for scoring purposes, an animal Displaced by A would also be scored as having been Approached by A.

Sniffs — Sniff refers to one animal placing its nose against, or within 1-inch of the body of a second animal. At such times, the vibrissae may be noticeably moved, and two animals may sniff at each other simultaneously.

Activity — A rough indicant of locomotor activity was obtained for each subject by counting the number of times that it crossed through the tunnel from one half of the cage to the other during each 30-min period.

Pauses near another animal — One animal stops and remains near (within approximately 4 in) or touching a second animal for at least 3-sec.

Miscellaneous social behavior — Instances of social interactions not scored in the matrices described above were also recorded for each session (e. g., biting, mounting, and grooming).

Results

The results are based upon a total of 109 h (i. e., 218 30-min sessions) of observation. Five of the eight triads were observed for a full 20-days (17 h each), while the remaining three (see below) were observed for a lesser period of time due to the death of one animal in each of these triads.

Social interactions within like-sexed triads

A question of special interest during the course of the observation was that of whether there are any sex differences in the degree of agonistic behavior in *A. cahirinus*. One method of investigating this question consists of com-

Table 1: Mean frequencies (per hour of observation) of occurrence of specific categories of social interactions for the single-sex triads

	Group			
	3 - ♂♂ I (Previously housed in heterosexual pairs)	3 - ♂♂ II (Isolated post-weaning)	3 - ♀♀ I (Previously housed in heterosexual pairs)	3 - ♀♀ II (Isolated post-weaning)
<u>Behavioral Category</u>				
Chases	12.0	14.8	46.6	20.7
Approaches	106.8	29.9	86.3	101.9
Displaces	44.7	15.4	60.3	65.2
Sniffs	50.8	22.5	43.5	34.9
Grooming	.3	.0	.0	.1
Two (or three) animals pause near each other	30.5	4.0	5.0	4.4
Bites	.4	.1	.0	1.0
Activity	135.5	127.8	228.6	158.4
<u>Displaces</u>				
Approaches	.42	.51	.70	.64

paring the frequencies of the various behavioral categories described above (especially Chases, Displaces and Bites) for the all-♂ vs. all-♀ triads. These data are presented in tabular form in Table 1, with the tabled frequencies representing the mean frequencies of occurrence of the various behaviors for the entire triad per hr of observation. A further gross indicant of the degree of agonistic behavior was calculated by dividing the number of displaces/hr for each triad by the mean number of approaches for that triad. These proportions are presented in the final row of Table 1.

As indicated in Table 1, in comparison to the two all-♂ triads, each of the all-♀ triads displayed a greater mean frequency of the following behavioral categories: Chasing, Displaces, and Activity. Also, the ratio of displaces/approaches was greater for each of the female triads than for either of the male triads.

Finally, whereas all of the animals in the two all-♂ triads survived the 20-day observation period, one of the six animals from the two female triads was found dead and partially cannibalized on the 18th day of observation. Since a large wound was noticed on the back of this ♀ one day prior to her death, it appears as though her death resulted from attacks by the other ♀♀ in that triad.

Agonistic interactions within mixed-sex triads

Each of the triads consisting of two ♀♀ and one ♂ was observed for the full 20-days (17 h of observation). However, for the triads made up of two ♂♂ and one ♀, the observations were curtailed due to the deaths of one of the ♂♂ in each group (on observation days 3 and 9 respectively). Therefore, the total observations for the two triads consisting of two ♂♂ and one ♀ were 2 h and 7 h respectively. In each instance, the death of one of the ♂♂ appeared to be due to attacks by the other members of the triad since large, fresh wounds were visible prior to finding the animal dead. Furthermore, within each of these triads, the single ♀ was the only animal ever observed to attack another.

For the triad in which one ♂ was found dead after 2 h the ♀ was observed to bite that ♂ twice, and bite the second ♂ on one occasion. While for the triad in which one of two ♂♂ was found dead after 7 h of observation, the ♀ had been observed to bite that ♂ a total of 8 times, and to bite the second ♂ a total of 11 times. It therefore appears that the large wounds and subsequent deaths of one of the ♂♂ in each of the triads made up of two ♂♂ and one ♀ were primarily due to attacks by the single ♀ member of these triads. Likewise, although very few occurrences of biting were observed in either of the triads consisting of two ♀♀ and one ♂, in each instance all of the biting was performed by one of the ♀♀ within each group. Thus, all observed biting within the four mixed-sex triads was performed by ♀♀, and in two instances, such attacks by ♀♀ appear to have been responsible for the deaths of male conspecifics. These data are consistent with the results from the like-sexed triads in which the sole death occurring during the observation period consisted of a ♀ who had been apparently killed by the other female members of her triad.

Correlational analysis of frequencies of behavior patterns

Within each of the 8 triads, the frequencies of occurrence of the 9 most common behavioral categories, along with frequency of success in obtaining and eating the piece of apple first dropped into the cage each day, were calcu-

Table 2: Matrix of mean Spearman's rank-order correlation coefficients (across all 8 triads) for 10 behavioral categories

	Eats Apple	Sniffed At	Sniffs	Activity	Displaced	Displaces	Approached	Approaches	Chased
Chases	+.30	+.50	+.36	+.38	-.75**	+.88**	-.81**	+.75**	-.77**
Chased	-.36	-.58	-.34	-.14	+.86**	-.77**	+.73*	-.61*	
Approaches	+.55*	+.44	+.61*	+.63*	-.56*	+.75**	-.81**		
Approached	-.52	-.69**	-.39	-.44	+.81**	-.81**			
Displaces	+.36*	+.63*	+.36	+.38	-.75**				
Displaced	-.42	-.75*	-.20	-.13					
Activity	+.52*	+.25	+.55*						
Sniffs	+.56*	+.48							
Sniffed at	+.58*								

* $p < .05$; ** $p < .01$.

lated for each animal. Low frequencies of occurrence of the other behavioral categories precluded meaningful correlational analyses of them. The categories *Chases*, *Approaches*, *Displaces* and *Sniffs* refer to behavioral categories described above; whereas the categories *Chased*, *Approached*, *Displaced* and *Sniffed at* refer to the focus (or target) of these categories of behavior. Each of the three animals within a triad was ranked according to the frequency with which it displayed the 10 behavioral categories, and these ranks were then intercorrelated using Spearman's coefficient of rank correlations. The mean values (for all 8 triads) of these rank correlations between the 10 behavioral categories are presented in matrix form in Table 2. These mean correlation coefficients were tested for significance by transforming the individual correlation coefficients for each of the 8 triads into Z scores and then computing t tests for single samples with 7 ($n-1$) degrees of freedom.

Miscellaneous observations

It is of especial interest that actual fighting, in which two animals simultaneously bite at each other, was briefly observed only four times during the 109 h of observation. All of the other occurrences of agonistic interactions (e.g., chasing, displaces, biting) were one-sided; with even such unilateral biting being relatively uncommon (mean frequency = 0.8/h across the 8 triads) and not differing greatly between triads (range = 0–2.7/h). Thus, in general, agonistic encounters tended to consist of a series of *Chases* or *Displaces* with few observed instances of physical contact (e.g., biting) being involved.

Unlike laboratory mice (*Mus musculus*) and rats (*Rattus norvegicus*), *A. cahirinus* were never observed to adopt a noticeable stereotyped submissive posture when attacked by a second animal. The attacked animal would usually simply move away from the aggressor, often emitting highpitched squeals when first attacked or bitten.

Throughout the observation period, there was a noticeable lack of overt sexual behavior such as mounting and copulation. In fact, the only mounting observed was homosexual mounting in one of the all-♂ triads. Within that triad, the ♂ most often seen chasing and displacing the others twice appeared to mount another ♂. In both instances, these mount attempts were very brief, with the mounted animal quickly running away from the other ♂.

II. Aggressive Behavior during Paired Encounters between male and female *A. cabirinus*

The results of the above observational study of adult *A. cabirinus* suggest that in comparison to ♂♂, ♀♀ may be more aggressive; and possibly even dominant over male conspecifics. However, the total number of animals of either sex that were observed did not allow for a direct statistical test of this hypothesis. It was therefore decided to investigate further any possible sex differences in aggressiveness in *Acomys c.* through a series of paired encounters between adult ♂♂ and ♀♀.

Methods

Subjects — A total of 64 adult (i.e., at least 90 days of age) spiny mice (32 of each sex) were used in this study. These animals had been born in the laboratory and housed in individual cages (7 × 11 × 5 in high) after being removed from their parents and siblings at 30 days of age.

Procedure — ♂♂ and ♀♀ were assigned to heterosexual pairs on a random basis. Each pair was tested once in the isolation-cage of the ♂, and once in the cage of the ♀. The order of the location for the two tests (i.e., male or female cage) was randomly determined for each pair, and there was an interval of at least two weeks between the first and second test. There were thus a total of 64 paired encounters (2 encounters each for the 32 pairs of animals) between adult male and female *Acomys*.

At the time of testing, one member of a pair was removed from its isolation cage and placed into the cage of the opposite-sexed member of that pair. The two animals were then observed until aggressive behavior occurred (i.e., biting, or one animal repeatedly chasing a retreating animal) or until a maximum of 10 min had elapsed. At this time, the animals were separated and again isolated in their respective cages until the second test.

Results

The observed instances of aggressive behavior consisted of one animal either biting the second animal or repeatedly pursuing the second animal, who appeared to be attempting to escape from its pursuer. No instances of reciprocal biting were observed during these tests.

The results of this experiment are summarized in Table 3. When tested in the home cage of the *female* member of the pair, the frequency of female aggression directed towards the ♂♂ was significantly greater than was the frequency of male aggression ($\chi^2 = 24.0$, d/f 1, $p < .001$). However, for the tests in the ♂♂ home-cages, no significant difference was found between the frequencies of male vs. female aggression.

A further statistical test compared the proportion of paired encounters in the ♀♀' cages which resulted in aggression (24/32) with the proportion of encounters in the ♂♂' cages during which aggression was observed (13/32). The results of this test indicate that aggressive behavior was significantly

Table 3: Results of paired encounters between male vs. female conspecifics
(i.e., number of such encounters resulting in overt aggression)

Aggressor	Location of paired encounter	
	♂'s home cage	♀'s home cage
♂	5	0
♀	8	24
Neither animal (i.e., no aggression)	19	8

more frequent in the isolation cages of the ♀♀ than in the cages of the ♂♂ ($z = 2.64$, $p < .01$, 2-tailed test).

Discussion

The results of the first study indicate that agonistic encounters between adult *Acomys c.* are characterized by frequent chases and physical displacements along with a relatively low frequency of actual physical contact including biting. A subordinate animal appears to attempt to move away from a superior or attacking animal — with there being no apparent appeasement gesture(s) functioning to reduce further aggression. Therefore, continued attack can be fatal to the subordinate animal as evidenced by the deaths (apparently resulting from attack) of animals in three of the triads.

Based upon the correlational analysis of frequencies of occurrence of various behavior patterns, an aggressive animal would be characterized as one that frequently Chases, Displaces, and Approaches other members of the triad, while being itself only infrequently Approached, Chased or Displaced by the others. On the other hand, subordinate animals are characterized as those who are frequently Chased, Approached, and Displaced, while infrequently Chasing, Displacing or Approaching others. It should be noted that an animal's success in obtaining the single apple piece when first dropped into the cage did not show a consistently high correlation with its relative frequency of performance of the directly aggressive behavioral categories. Thus, while the correlation between obtaining the apple and Displaces was *marginally* significant, the absolute value of the correlation was relatively low (.36). Furthermore, no significant correlation was found between rankings of Chases (the most obviously aggressive behavioral category in the correlation matrix) and success in obtaining the apple piece. On the other hand, the significant positive correlation between Activity and Apple Eating suggests that the most active animal in the triad may be quite successful in locating and obtaining the apple regardless of his level of agonistic behavior outside of that context. In a related study, a lack of correlation between behavior in a food competition task and other measures of aggression (e. g., shock-elicited fighting, and mouse killing) was found in septal-lesioned rats (MICZEK and GROSSMAN 1972). Likewise, in pigtail monkeys, priority of access to incentives (e. g., entry into a feeding area, or feeding order) "may correlate poorly with the more direct measure of status as defined by 'pecking order'" (BERNSTEIN 1969). Thus, as recently pointed out by HINDE (1974), across vertebrate phyla, "access to food depends on subtleties of behavior, such as opportunism, which cannot be assessed in terms of a unitary variable" (such as agonistic behavior or dominance).

The results from both the observational study and the series of paired encounters between ♂♂ and ♀♀ suggest that adult female *A. cahirinus* tested in their home areas are more aggressive than are the ♂♂. In the all-♀ triads, displays of such agonistic behaviors as Chasing and Displacing were more frequent than in the all-♂ triads. Furthermore, in the mixed-sex triads, all observed occurrences of biting were performed by ♀♀, who appeared to be at least partially responsible for the deaths of two ♂♂. During the paired encounters, the ♀♀ attacked the ♂♂ significantly more often than vice versa when tested in the home cages of the ♀♀. On the other hand, no differences were found between frequencies of male vs. female aggression when tested in the home cages of the ♂♂. It therefore appears that familiarity with

the environment tends to increase the likelihood of ♀ aggression being directed towards a conspecific of the opposite sex. Similar instances of heightened aggression on the part of the "home cage animal" (♂♂) have been reported for laboratory mice (UHRICH 1938, JONES and NOWELL 1973). These findings are, of course, analogous to the frequently reported increased probability of victory on the part of the animal (of many species) on whose territory an intraspecific conflict occurs.

Although the greater aggressiveness of female as compared to male *A. cahirinus* does not conform to the typical rodent pattern, such sex differences in agonistic behavior (and dominance) are not entirely unique to this species. In the golden hamster (*Mesocricetus auratus*), for example unreceptive ♀♀ are highly aggressive and typically dominant over male conspecifics (DIETERLEN 1959; PAYNE and SWANSON 1970, 1972). However, when in estrus, female hamsters display much less aggression towards conspecifics of either sex (PAYNE and SWANSON 1970). Further studies have revealed that specific endocrine manipulations interact in a complex manner with the sex of the opponent to influence the degree of overt aggression displayed by female hamsters (cf. PAYNE and SWANSON 1972; VANDENBERGH 1971).

The similarity between hamsters and *A. cahirinus* in regards to female aggressiveness begs the question of whether the endocrine basis of aggression is likewise similar in these two species. In any event, further study of the agonistic behavior of *Acomys cahirinus* (especially in regards to the endocrine control thereof) would certainly seem warranted.

Summary

Two studies of the agonistic behavior of *Acomys cahirinus* (spiny-mouse) were reported. The first consisted of a series of observations of eight groups consisting of three adult animals each. Although fighting and biting were rarely observed, instances of chasing and physical displacement were quite common. The frequencies of these latter two behavioral categories were greater within the groups made up of three ♀♀ than within the all ♂♂ groups. In heterosexual groups, ♀♀ were observed to bite the other members of the triad on several occasions; however, biting by ♂♂ was never observed in these same triads. Three animals were found dead during the course of the observations — with their deaths apparently resulting from attacks by female cage-mates. A correlation matrix for the frequencies of occurrence of various behavioral categories was also included.

In the second study, the outcomes of paired encounters between male and female *Acomys c.* were recorded. For the encounters conducted in the home cages of the ♀♀, the frequency of female aggressive behavior directed towards ♂♂ was significantly greater than that of the ♂♂ directed towards ♀♀. Likewise, more instances of aggression were observed in the cages of the ♀♀ than in the cages of the ♂♂. The results of these two studies indicate that unlike most species of rodents studied to date, in *Acomys cahirinus*, ♀♀ in their home area are typically more aggressive than, and dominant over, male conspecifics.

Zusammenfassung

Beobachtungen an 8 Gruppen von jeweils drei erwachsenen Stachelmäusen (*Acomys cahirinus*) zeigten ziemlich häufiges Vorkommen von Hetzjagden

und körperlichem Abdrängen, während Kämpfe und Bisse selten beobachtet wurden. Das Verhältnis der häufigsten Verhaltenskategorien zueinander wurde in einer Korrelationsmatrix dargestellt.

In einer zweiten Studie getestete ♀♀ dominierten in ihrem Heimkäfig über ♂♂. Im Heimkäfig der ♂♂ wurde jedoch kein Unterschied in der Häufigkeit der Aggressionen von ♀ gegenüber ♂ beobachtet. In den Käfigen der ♀♀ wurden mehr Fälle von Aggression beobachtet als in den Käfigen der ♂♂.

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